

Supriya Khadka

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Education

Ph.D. in Information Technology

Aug 2026 – Present

George Mason University, United States

Advisor: Dr. Sanchari Das

Master of Science by Research in Artificial Intelligence

Sep 2024 – Jan 2026

Coventry University, United Kingdom

Advisors: Dr. Xiaorui Jiang, Prof. Vasile Palade

Bachelor of Engineering, Computer Engineering

Nov 2018 – Apr 2023

Pulchowk Campus, Tribhuvan University, Nepal

Research Experience

Graduate Researcher, DAS Lab, George Mason University

Oct 2025 - present

Research Area: Usable Privacy and Security

Advisor: Dr. Sanchari Das

- Investigating usable privacy and security for AI agents and companions, with a current focus on systems that handle sensitive health information.
- Conducting systematic and empirical HCI research on privacy, trust, authentication, and user control across AI and emerging technologies.

Visiting Researcher, CHAI Lab, Northeastern University

Oct 2025 - July 2026

Research Area: Empathy in Conversational Agents

Advisor: Dr. Smit Desai

- Conducted a systematic review of 89 studies on empathy in conversational agents, synthesizing trends in their design, evaluation, and applications.
- Developed a conceptual framework connecting interaction contexts, empathy design strategies, and evaluation methods in conversational agents.

Postgraduate Researcher, Coventry University

Sept 2024 - Jan 2026

Project Title: Lowering the Cost of Healthcare by Clinical Coding with LLMs

Advisors: Dr. Xiaorui Jiang, Prof. Vasile Palade

- Developed and evaluated an LLM-assisted clinical coding pipeline spanning evidence extraction, code prediction, and verification.
- Assessed dataset quality, coding errors, model performance, and cost-effective adaptation strategies in collaboration with professional clinical coders.

Advisors: Dr. Basanta Joshi (CARD), Dr. Bishesh Khanal (NAAMII)

- Developed a Nepali text-to-speech model and mobile application, leading to a SIGUL publication, and publicly released a comprehensive Nepali speech dataset through [OpenSLR](#).
- Analysed and curated Nepali datasets for sentiment analysis, text-to-speech, and machine translation, documenting their characteristics and limitations.

Teaching Experience

Graduate Teaching Assistant, George Mason University

Aug 2026 - present

- Supporting two sections of IT 304 (IT and the Global Economy), serving a total of 76 students.
- Assisting with course administration, grading assignments and quizzes, and facilitating research activities and group discussions.

Undergraduate Research Coordinator, Sunway College Kathmandu

Jan 2026 - Jun 2026

- Taught a semester-long course on research methods and academic writing, supervised Individual Honours Projects, and led a weekly paper-reading club.
- Mentored 18 undergraduate students on individual and group research projects, resulting in six publications at peer-reviewed international workshops and conferences.

Technical Lead and Facilitator, Women Leaders in Technology

Sept 2020 - Aug 2024

- Progressed from fellow to Technical Facilitator across four fellowship cohorts, teaching Git, JavaScript, Express, and MongoDB to over 60 women in technology in Nepal.
- Promoted to the Technical Lead in the latest batch, responsible for developing the curriculum, training junior facilitators, overseeing all technical sessions and projects, and teaching approximately half of the classes throughout the fellowship.

Industry Experience

Conversational AI Engineer, Diyo.AI

May 2023 - Aug 2024

- Built multilingual chatbots for sexual, reproductive, and maternal health, safe abortion, e-governance, and sales.
- Evaluated and integrated GPT-3.5, Llama 2, BLOOM, and Okapi and curated low-resource NLP datasets to improve conversational AI for Nepali and other languages of Nepal.

Selected Publications

Khadka, Supriya, Xiaorui Jiang, and Vasile Palade. “LLM-Assisted Clinical Coding Audit through an Interpretable Coding Pipeline”. *International Journal of Medical Informatics*, vol. 220, 2026, p. 106605. <https://doi.org/10.1016/j.ijmedinf.2026.106605>.

Khadka, Supriya, and Smit Desai. “Empathy through the Lens of Conversational Agents: A Systematic Review”. *Proceedings of the ACM Conversational User Interfaces Conference (CUI '26)*. ACM, 2026, pp. 1–22, <https://doi.org/10.1145/3816046.3816220>. **Honorable Mention Award**.

- Khadka, Supriya**, and Sanchari Das. “Privacy-Preserving Compliance on Public Ledgers via Selective Disclosure Authorization Schemes”. *Proceedings of the 23rd International Conference on Security and Cryptography (SECRYPT 2026)*. SCITEPRESS, 2026.
- Khadka, Supriya**, and Sanchari Das. “SoK: Understanding the Pedagogical, Ethical, and Privacy Challenges in Adopting XR for Early Childhood Education”. *Proceedings of the Augmented Humans International Conference 2026*. ACM, 2026, pp. 357–75, <https://doi.org/10.1145/3795011.3795019>.
- Khadka, Supriya**, and Bijayan Bhattarai. “Gender Bias in Nepali-English Machine Translation: A Comparison of LLMs and Existing MT Systems”. *Proceedings of the 6th Workshop on Gender Bias in Natural Language Processing (GeBNLP)*. Association for Computational Linguistics, Aug. 2025, pp. 75–82, <https://doi.org/10.18653/v1/2025.gebnlp-1.6>.
- Khadka, Supriya**, Xiaorui Jiang, and Vasile Palade. “Automated Clinical Coding with Multiclass ICD Prediction”. *2025 International Conference on Machine Learning and Applications (ICMLA)*. 2025, pp. 1295–300, <https://doi.org/10.1109/ICMLA66185.2025.00198>.
- Khadka, Supriya**, Ranju G.C., Prabin Paudel, Rahul Shah, and Basanta Joshi. “Nepali Text-to-Speech Synthesis using Tacotron2 for Melspectrogram Generation”. *2nd Annual Meeting of the ELRA/ISCA SIGUL*. 2023, pp. 73–77, <https://doi.org/10.21437/SIGUL.2023-16>.

Peer-Reviewed Extended Abstracts and Posters

- Khadka, Supriya**, and Sanchari Das. “Grant, Verify, Revoke: A User-Centric Pattern for Blockchain Compliance”. *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. ACM, 2026, pp. 1–6, <https://doi.org/10.1145/3772363.3798823>.
- Khadka, Supriya**, Dhiman Goswami, and Sanchari Das. Poster: Privacy-Preserving Compliance Checks on Ethereum via Selective Disclosure. 2026. www.ndss-symposium.org/wp-content/uploads/ndss26-poster-32.pdf.
- Khadka, Supriya**, and Sanchari Das. “XR Design Framework for Early Childhood Education”. *2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. 2026, pp. 1253–54, <https://doi.org/10.1109/VRW70859.2026.00301>.

Honors

Scholarships and Fellowships

- British Council Women in STEM Scholarship, 2024
- Ncell Excellence Scholarship - Yearwise Top Student, 2019 - 2023 [Press Releases [1](#), [2](#), [3](#), [4](#)]
- Fusemachines AI Fellowship, 2022
- 3rd Annual Nepal AI School - Full Scholarship, 2021
- Grace Hopper Conference Scholarship, 2021
- Women Leaders in Technology Fellowship, 2020

Awards

- Honorable Mention, ACM CUI 2026
- Best Poster, 3rd Annual CSMM Conference, 2025
- GritFeat AI Health Hackathon, Runner Up, 2023
- Hack-A-Week Winner (Category - FinTech), 2022
- BPKHIS and GDHUB Hackathon for Healthcare, Runner Up 2022
- OpenIMIS Datathon, 2nd Runner Up, 2022

Academic Service and Mentorship

Program Committee and Peer Review

- **Program Committee:** Second Workshop on Challenges in Processing South Asian Languages (CHiPSAL 2026), LREC 2026
- **Peer Reviewer:** GeBNLP at AACL 2026; ACM CUI 2026; ACM CSCW 2026 Posters; CSMM 2025

Conference Volunteering

- 14th ACM Collective Intelligence Conference (HCOMP-CI 2026)
- 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)

Mentorship

- **Undergraduate Students (6):** Lakshmi Katrapati (GMU), Samikshya Dhamala (Sunway), Raulnak Regmi (Sunway), Suraj Patel (Sunway), Kashish Dhami (Sunway), and Smreeta Shrestha (WLiT)
- **High School Student (1):** Sharani Chebrolu (GMU ASSIP)

Additional Leadership

- Planning Coordinator, GitHub Field Day Nepal, 2023–2024
- Technical Manager, Pulchowk Girls, 2020–2022

Selected Talks and Presentations

- **Poster:** From Evidence to Codes: Making Automated Clinical Coding Explainable. *3rd Annual CSMM Conference*, 2025.
- **Oral Presentation:** From Evidence to Codes: A Two-Stage Approach for Explainable Clinical Coding. *Researcher Capability and Development Conference (RCAD)*, 2025.
- **Keynote:** Navigating Gender Biases in AI: Is It Us or Is It the Data? A Perspective on Nepal and Beyond. *Women in Data Conference*, 2024. [Press coverage](#)
- **Poster and Oral Presentation:** Shruti: Nepali Text-to-Speech Synthesis Using Tacotron2 for Mel-Spectrogram Generation. *Young Women Leading the Way Showcase*, 2023.